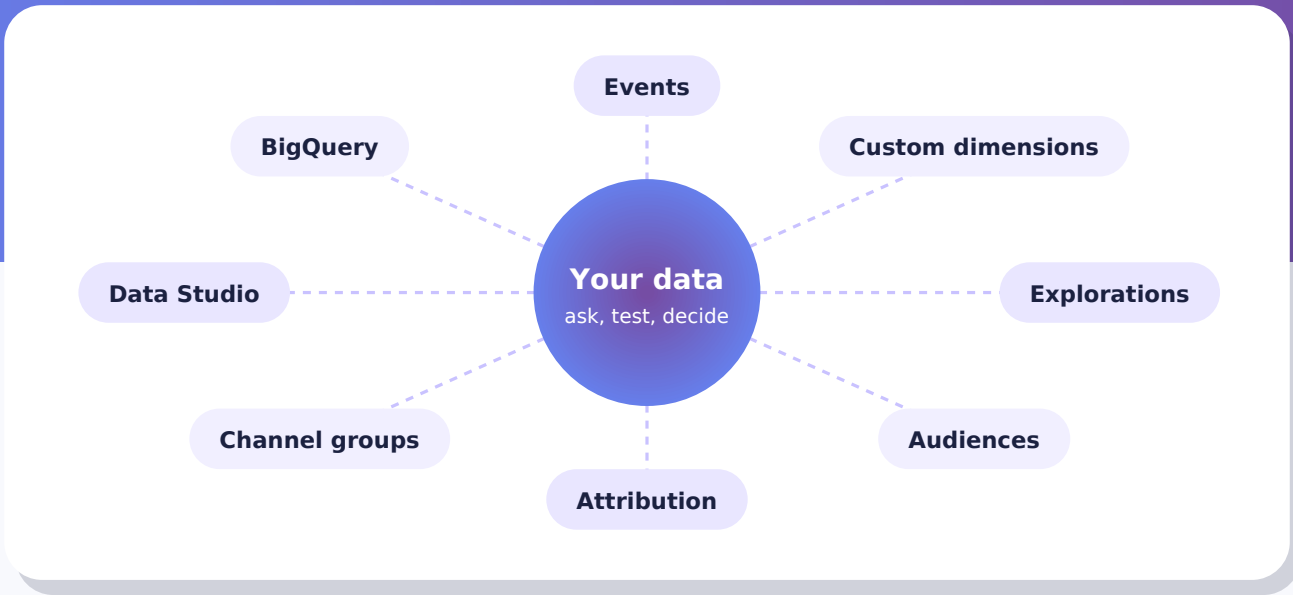
WEEK 11 · DIGITAL MARKETING & AI BOOTCAMP

Advanced Analytics & Data

Data mastery with GA4: events, explorations, attribution and Data Studio dashboards

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PART A

Foundations

From reading reports to real analysis:
the mindset, the GA4 data model
and why scope matters

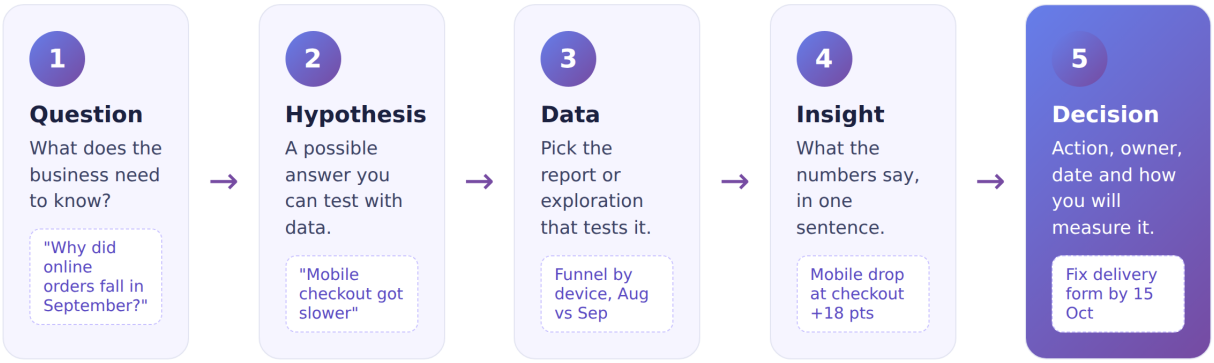
SECTION 1

From reports to analysis: the analyst’s mindset

In Week 2 you installed Google Analytics 4 (GA4), set up Google Tag Manager (GTM), tagged links with UTMs, created key events and built a first Data Studio report. This week you go deeper: you learn how GA4 stores data, how to collect exactly what the business needs, and how to answer real questions with explorations, attribution and dashboards.

The analysis mindset: from question to decision

Start with a business question, not with a report. Every analysis should end with an action.



Every analysis follows the same five steps. Most beginners jump straight to "Data" and get lost.

Reporting vs analysis

	Reporting	Analysis
Question	"What happened?"	"Why did it happen and what should we do?"
Output	Dashboards, monthly numbers	An insight and a recommendation
Tools	GA4 standard reports, Data Studio	Explorations, segments, attribution, BigQuery, Excel
Rhythm	Weekly or monthly, always the same	When a question or a change appears
Risk	Numbers nobody reads	Analysis that never becomes an action

Good business questions

A good question is specific, linked to money or customers, and answerable with data you have. Compare:

Vague	Answerable
"How is the website doing?"	"Which landing pages bring paid social visitors who order, and which ones lose them?"
"Is Instagram working?"	"How many orders start with an Instagram visit, even if the last click is Google or email?"
"Why are sales down?"	"Did the share of mobile visitors who reach checkout fall after the new delivery form on 2 September?"
"Do people come back?"	"What share of customers acquired in August visited again within 4 weeks, by channel?"

The four rules of an analyst

1. **Always compare.** A number alone means little: compare with last period, another channel, another device or a target.
2. **Segment before you conclude.** Averages hide the story. "Engagement rate 58%" could be 75% on desktop and 40% on mobile.
3. **Know where the number comes from.** Which scope, which attribution model, which date range, is consent or thresholding affecting it? (Sections 3, 18 and 27.)
4. **End with a decision.** Action, owner, date and the metric that will show if it worked. Data that does not change a decision is decoration (Week 2, Section 25).

> **This week's deliverable:** an **Analytics Report**, a custom GA4 dashboard in Data Studio with at least 5 metrics, plus an attribution analysis with your recommendations. Section 29 gives the full brief.

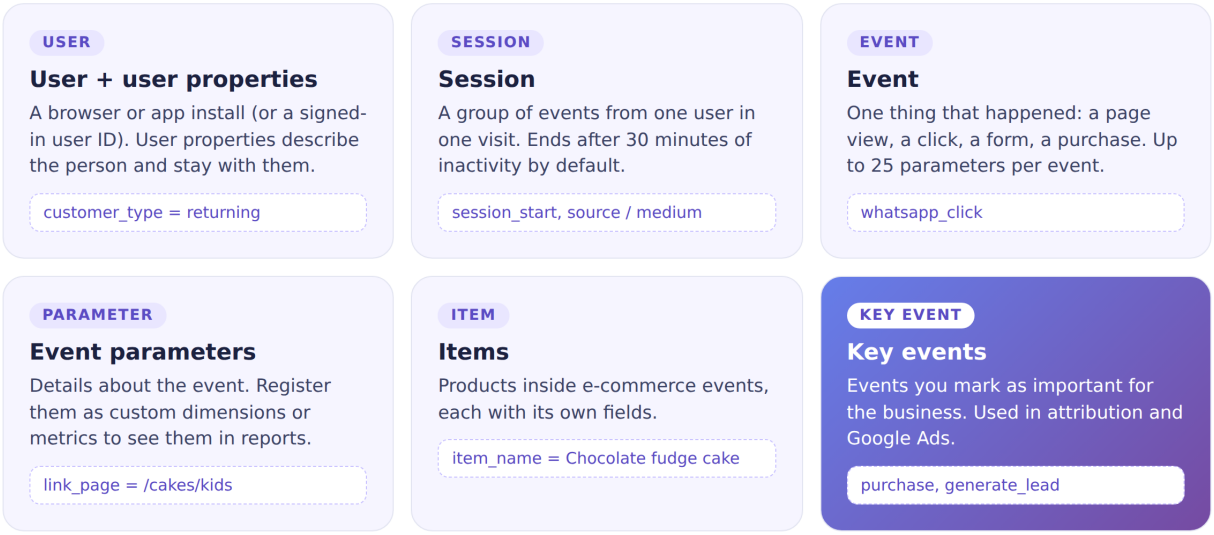
SECTION 2

The GA4 data model: events, parameters, user properties and items

To customise GA4 you need to understand how it stores data. Universal Analytics (the old version) was built around sessions and page views. GA4 is built around events: every interaction is an event, and everything else is calculated from events.

The GA4 data model in one picture

Everything is an event. Events carry parameters; users carry properties; e-commerce events carry items.



The building blocks

Element	What it is	Lakaz Gato example
Event	An interaction, with a name (lowercase, underscores, up to 40 characters)	page_view, whatsapp_click, purchase
Event parameter	Extra detail sent with the event. Up to 25 parameters per event	link_page, form_name, value, currency
User property	An attribute of the user, stored with all their later events	customer_type = returning; preferred_language = fr
User ID	Your own ID for a signed-in customer (never an email or phone number)	Customer number 10482 from the shop
Item	A product inside an e-commerce event, with item parameters	item_name, item_category, price, quantity
Key event	An event you marked as important	purchase, generate_lead, whatsapp_click

Four kinds of events

Kind	Who defines it	Examples	Action needed
Automatically collected	GA4	first_visit, session_start, user_engagement	None
Enhanced measurement	GA4 (switch on/off per stream)	page_view, scroll (90%), click (outbound), view_search_results, file_download, form_start, form_submit, video_start	Check each option in the web stream settings
Recommended	Google names them, you send them	purchase, add_to_cart, generate_lead, sign_up, login	Use Google’s exact names and parameters (Section 8)
Custom	You	whatsapp_click, order_form_submit, menu_download	Create in GTM or GA4 (Section 7)

Why parameters matter

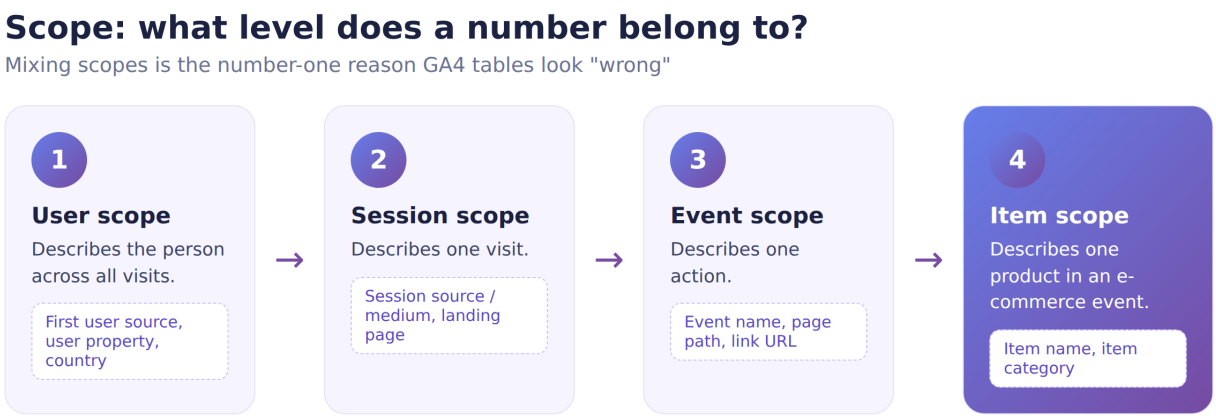
An event name tells you *what* happened; parameters tell you *where*, *which* and *how much*. "whatsapp_click" alone tells you 140 people clicked WhatsApp. Add the parameter **link_page** and you learn that 90 of them clicked on the wedding cakes page, which tells you where the hot leads are. But GA4 will not show most parameters in reports until you **register them as custom dimensions or metrics** (Section 9).

> **Personal data never goes into GA4.** Google’s policies forbid sending names, email addresses, phone numbers or similar information in events, parameters, user IDs or page URLs. Check that your forms do not put the email address in the thank-you page URL (for example /merci?email=…).

SECTION 3

Scope: why your GA4 numbers do not always add up

Every dimension and metric in GA4 belongs to a scope: user, session, event or item. When you combine dimensions and metrics from different scopes in one table, numbers can look strange. Understanding scope saves hours of confusion.



Three classic surprises

What you see	Why	What to do
The total of a table is lower than the sum of its rows	A user who visited from Facebook and later from Google counts once in the total but once in each row	Normal for user metrics. Use sessions or events when you need rows that add up
"First user source" and "Session source" disagree	First user = how the person first arrived (user scope). Session source = how this visit started (session scope)	Use first user dimensions for acquisition of new customers, session dimensions for campaign performance
Item revenue does not match total revenue	Item revenue is item scope; purchase revenue also includes shipping and tax	Compare like with like: item reports for products, purchase revenue for totals

Matching dimensions and metrics

- **User dimensions** (First user default channel group, country, user properties) go with **user metrics** (Total users, New users, Active users).
- **Session dimensions** (Session source / medium, Session campaign, Landing page) go with **session metrics** (Sessions, Engagement rate, Session key event rate).
- **Event dimensions** (Event name, Page path, custom event dimensions) go with **event metrics** (Event count, Key events).
- **Item dimensions** (Item name, Item category) go with **item metrics** (Items purchased, Item revenue).

> **Tip:** if a table looks wrong, ask "what scope is each column?" before you blame tracking. Mixing a session dimension with a user metric is allowed, but the totals will not add up.

Sessions, users and key events: quick reminders

A session ends after 30 minutes of inactivity (you can change this in the web stream tag settings). A **user** is a browser or device, unless you send a User ID for signed-in customers. The same person on a phone and a laptop is two users unless GA4 can link them. **Key events** are counted every time the event happens, so if one person submits the form twice, you get two key events. That is why "Session key event rate" and "User key event rate" exist: they count sessions or users with at least one key event.

SECTION 4

Analytics maturity: where Mauritian businesses stand

Most small businesses in Mauritius are at the start of the journey: GA4 is installed, someone looks at the number of users from time to time, and decisions are made on instinct. You do not need to reach the top level, but each step up brings better decisions.

Level	What it looks like	Next step
1 · Installed	GA4 tag on the site, default reports, nobody owns it	Settings from Week 2; 1–3 key events
2 · Tracked	Key events and UTMs work; monthly check of users and channels	Measurement plan; custom events with parameters
3 · Reported	A Data Studio dashboard everyone uses; monthly review meeting	Explorations, segments and audiences
4 · Analysed	Questions answered with funnels, cohorts, attribution; tests run	Attribution and channel strategy; BigQuery for history
5 · Data-driven	Data from GA4, ads, CRM and sales joined; budgets set with evidence	Experiments, incrementality, modelling

What is different in Mauritius

- **WhatsApp closes many sales.** A visitor reads the menu on the website, then orders on WhatsApp. GA4 sees the click on the WhatsApp button, not the sale. You need a click event (Section 7) and, ideally, a simple way to match WhatsApp orders back to campaigns (a code in the pre-filled message, Section 20).
- **Small volumes.** A local business may have 3,000 sessions a month, not 3 million. Some features need more data: data-driven attribution works best with many conversions, and thresholds can hide rows with few users (Section 27). Use longer date ranges and simpler segments.
- **Mobile first.** Most visits come from phones. Always break down by device category.
- **Mixed offline and online.** Many orders are paid in the shop, by bank transfer or by mobile payment. Keep a simple sales log (Excel or CRM) with the source of each order so you can reconcile with GA4 (Section 26).
- **Tourists and diaspora.** Hotels, guesthouses and online shops see visitors from France, Réunion, the UK, South Africa and India. Country and language dimensions matter, and some visitors are in the EU, where the GDPR (General Data Protection Regulation) applies.

> **In Mauritius:** a 12-room guesthouse, a gym or a cake shop can reach level 3 or 4 in a few weeks with free tools: GA4, GTM, Data Studio and a Google Sheet. The barrier is not budget, it is the habit of asking questions and acting on the answers.

PART B

Plan and collect

Measurement plan, dataLayer, custom events in GTM, e-commerce events, custom definitions and consent

SECTION 5

The measurement plan and tracking specification

A measurement plan is a short document that links business objectives to the exact data GA4 must collect. It stops you tracking "everything" (and understanding nothing), and it is the brief a developer or agency needs to set up tracking correctly.

The measurement plan: from business goals to events

Work from the top down. Never start by tracking "everything".



Build it in five steps

1. **Objectives:** write 1–3 business objectives with the owner (Week 1 SMART objectives).
2. **KPIs:** for each objective, 2–4 KPIs (key performance indicators) with a target: "60 online orders a month", "session key event rate 3%".
3. **Events:** list the events needed to calculate each KPI, with their parameters. Prefer recommended events (Section 8).
4. **Dimensions:** how you want to slice the KPIs: channel, device, landing page, product category, customer type.
5. **Owner and test:** who builds each event, how it is tested, and when it was last checked.

Tracking specification (example)

Event name	Fires when	Parameters	Key event?	Method
whatsapp_click	Click on any wa.me or api.whatsapp.com link	link_page, link_text, button_location	Yes	GTM click trigger
order_form_submit	Order form sent successfully	form_name, cake_type, delivery_area	Yes	dataLayer push + GTM
generate_lead	Wedding quote form sent	lead_source = wedding_form, value	Yes	dataLayer push + GTM
purchase	Online payment confirmed	transaction_id, value, currency = MUR, items	Yes	Shop integration
scroll_depth	User reaches 25, 50, 75 or 90% of a page	percent_scrolled	No	GTM scroll trigger
menu_download	Menu PDF downloaded	file_name	No	Enhanced measurement (file_download)

Naming rules

- lowercase with underscores: **order_form_submit**, never "Order Form Submit" or "orderFormSubmit". GA4 names are case-sensitive, so "Purchase" and "purchase" are two different events.
- Verb or object first, consistently: whatsapp_click, phone_click, map_click.
- Do not use reserved names or prefixes (for example names starting with "google_", "ga_" or "firebase_").
- Put the detail in parameters, not in the event name: one event **form_submit** with form_name = contact / quote / newsletter, not three events.

> **Tip:** keep the plan in a shared Google Sheet with tabs for objectives, events, custom definitions and a change log. When something breaks after a website update, the change log tells you what changed and when.

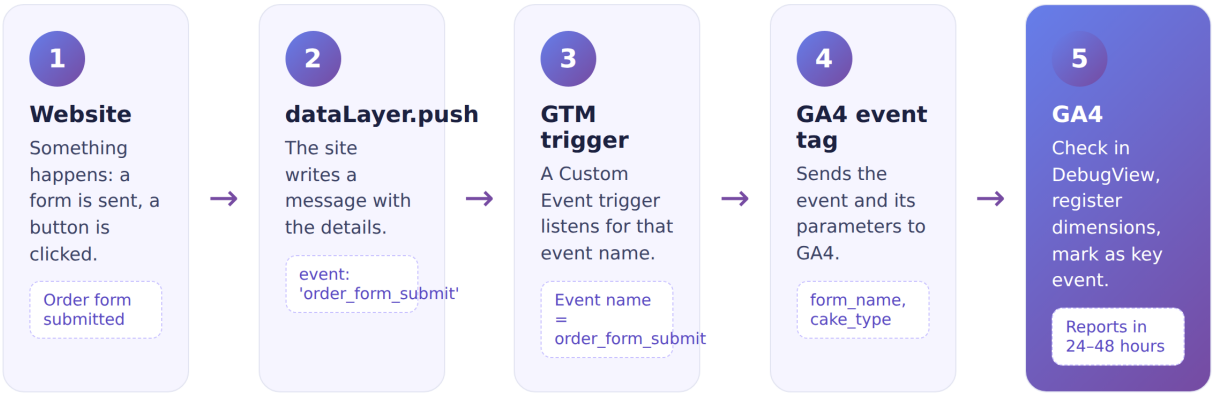
SECTION 6

The dataLayer explained

The dataLayer is a JavaScript list on your web pages where the website writes messages for Google Tag Manager to read. It is the most reliable way to track actions that clicks alone cannot capture, such as a successful form submission or a completed booking.

How a custom event travels from your website to GA4

The dataLayer is a message board between your website and Google Tag Manager



What a dataLayer push looks like

When the order form is sent successfully, the developer (or a plugin) adds this code to the page. You do not need to write it yourself, but you must be able to read it and brief a developer:

```
window.dataLayer = window.dataLayer || [];
window.dataLayer.push({
  event: 'order_form_submit',
  form_name: 'cake_order',
  cake_type: 'birthday',
  delivery_area: 'plaines_wilhems'
});
```

event is the name GTM listens for. The other keys are values GTM can read with **Data Layer Variables** and send to GA4 as event parameters.

Click trigger or dataLayer?

Method	Good for	Weakness
GTM click trigger	Links and buttons (WhatsApp, phone, map, "Book now")	Breaks if the button text or CSS class changes
GTM form submission trigger	Simple HTML forms	Does not work with many form plugins; may fire even if the form has errors
Thank-you page view	Forms that redirect to /merci or /thank-you	Counts again if the page is reloaded or bookmarked
dataLayer push	Successful submissions, bookings, purchases, anything with details	Needs a developer or plugin once

Where dataLayer events come from

- **E-commerce platforms:** Shopify, WooCommerce (with a GTM plugin) and most booking engines push e-commerce events automatically.
- **Form plugins:** many WordPress form plugins (Contact Form 7, Gravity Forms, WPForms...) can push a dataLayer event on success, directly or with a small add-on.
- **Your developer:** give them the tracking specification from Section 5. One hour of developer time saves months of bad data.

> **Tip:** open your website, press F12 (developer tools), go to Console and type **dataLayer** then Enter. You will see every message pushed so far. GTM Preview mode (Tag Assistant) shows the same list step by step.

SECTION 7

Custom events with Google Tag Manager, step by step

In Week 2 you tracked WhatsApp clicks with a simple click trigger. Here you build three events properly, with parameters and testing: a WhatsApp click, a successful form submission from the dataLayer, and scroll depth.

Before you start

☐ GTM installed on every page, GA4 Google tag firing on Initialization – All Pages (Week 2, Section 23)

☐ Built-in variables enabled: Click URL, Click Text, Click Classes, Page Path, Scroll Depth Threshold, Form ID

☐ GA4 DebugView open in another tab (Admin → DebugView)

Event 1 · whatsapp_click with parameters

1. **Trigger:** Triggers → New → Click – Just Links → Some Link Clicks → Click URL matches RegEx **wa\.me|api\.whatsapp\.com**. Name it "Click – WhatsApp".
2. **Tag:** Tags → New → Google Analytics: GA4 Event → Measurement ID (or the Google tag) → Event name **whatsapp_click**.
3. **Parameters:** add link_page = {{Page Path}}, link_text = {{Click Text}}, and button_location = a lookup of {{Click Classes}} (for example "header", "floating", "product") if your buttons have classes.
4. **Preview:** click Preview, open the site, click two different WhatsApp buttons. In Tag Assistant check that the tag fired once per click, with the right parameters.
5. **DebugView:** in GA4 the event appears with its parameters within seconds.
6. **Publish** with a clear version name: "v12 – whatsapp_click with parameters".

Event 2 · order_form_submit from the dataLayer

1. **Variables:** Variables → New → Data Layer Variable → name **form_name** (repeat for cake_type and delivery_area). Name them "DLV – form_name" and so on.
2. **Trigger:** Triggers → New → Custom Event → Event name **order_form_submit**.
3. **Tag:** GA4 Event → Event name order_form_submit → parameters form_name = {{DLV – form_name}}, cake_type = {{DLV – cake_type}}, delivery_area = {{DLV – delivery_area}}.

4. **Test** with a real form submission in Preview mode. Submit once with an error (empty field): the event must **not** fire. Then submit correctly: it fires once.

Event 3 · scroll_depth on long pages

Enhanced measurement only records a scroll at 90%. For blog posts and long sales pages you may want 25, 50, 75 and 90%. First switch off the enhanced measurement scroll option (to avoid two scroll events), then:

1. **Trigger:** Scroll Depth → Vertical scroll depths → Percentages **25,50,75,90** → Some pages → Page Path contains /blog (or All pages).
2. **Tag:** GA4 Event → Event name **scroll_depth** → parameter percent_scrolled = {{Scroll Depth Threshold}}.
3. **Register** percent_scrolled as an event-scoped custom dimension (Section 9) so you can use it in reports.

Then, in GA4

- Mark **whatsapp_click** and **order_form_submit** as key events (Admin → Key events, or star them in Admin → Events once they appear).
- Register the parameters you want in reports as custom dimensions (Section 9).
- Add the events to your measurement plan with the date and GTM version.

> **Avoid duplicates.** If you also created whatsapp_click in GA4 with "Create event" in Week 2, delete one of the two methods. Otherwise every click is counted twice and your key event rate doubles.

SECTION 8

Recommended events for e-commerce and lead generation

Google publishes a list of recommended events with exact names and parameters. If you use them, GA4 fills its ready-made reports (Monetization, Ecommerce purchases, funnels) automatically. If you invent your own names, those reports stay empty.

Online sales: the e-commerce journey

Step	Recommended event	Key parameters
See a list of products	view_item_list, select_item	item_list_name, items
See a promotion	view_promotion, select_promotion	promotion_name, items
See a product page	view_item	currency, value, items
Add to cart / wishlist	add_to_cart, add_to_wishlist, remove_from_cart, view_cart	currency, value, items
Checkout	begin_checkout, add_shipping_info, add_payment_info	currency, value, coupon, shipping_tier, payment_type, items
Purchase	purchase	transaction_id, value, currency, tax, shipping, coupon, items
Refund	refund	transaction_id, value, currency (items for a partial refund)

The **items** array holds one entry per product: item_id, item_name, item_category, price, quantity and more. Always send **currency** ("MUR") with value, otherwise GA4 cannot record revenue, and always send a unique **transaction_id** so a reloaded confirmation page is not counted twice.

Lead generation

For services (B2B, training, weddings, real estate), Google also recommends lead events that follow the sales process:

Event	Use it when
generate_lead	A form, quote request or call-back request is submitted
qualify_lead / disqualify_lead	Your team decides the lead fits (or not)
working_lead	A salesperson is in contact with the lead
close_convert_lead / close_unconvert_lead	The lead became a customer (or was lost)

The last four usually happen in your CRM (Week 10), not on the website. Advanced set-ups send them to GA4 later with the Measurement Protocol or Google's offline import tools, so you can see which channels bring leads that actually buy.

> **In Mauritius:** few local businesses send correct e-commerce events. If you sell online, test one real order every month with a small amount: check that purchase arrives once, with the right value in rupees, the right items and a transaction ID that matches your shop system.

SECTION 9

Custom dimensions and metrics

GA4 collects many parameters, but only a few appear in reports automatically. To use your own parameters (link_page, cake_type, customer_type...) as dimensions or metrics, you must register them as custom definitions.

Register a custom dimension

1. Make sure the parameter is being sent (you see it in DebugView).
2. Admin → Data display → **Custom definitions** → Create custom dimension.
3. Dimension name: a readable name, for example "Cake type".
4. Scope: **Event** for event parameters, **User** for user properties, **Item** for item parameters.
5. Event parameter: type the parameter name exactly as sent (cake_type). Save.
6. Wait: data appears from now on, usually within 24–48 hours. **Custom dimensions are not retroactive**: register early.

Limits for a standard (free) GA4 property

Custom definition	Standard property	Analytics 360
Event-scoped custom dimensions	50	125
User-scoped custom dimensions	25	100
Item-scoped custom dimensions	10	25
Custom metrics	50	125
Calculated metrics	5	50

Source: Google Analytics Help, "Dimensions and metrics" limits, checked October 2026. Admin → Custom definitions → Quota information shows how many you have used. After archiving a dimension you may have to wait about 48 hours before the slot can be reused.

Custom metrics and calculated metrics

- **Custom metric:** a numeric event parameter you want to add up, for example **deposit_amount** on a booking event or **guests** on a table reservation. Choose the unit (standard, currency, distance, time).
- **Calculated metric:** a formula on existing metrics, created in Admin → Custom definitions → Calculated metrics. Examples: Revenue per session = Purchase revenue ÷ Sessions; WhatsApp clicks per 100 sessions.

Good and bad custom dimensions

Good (few possible values)	Bad (high cardinality)
cake_type (birthday, wedding, cupcakes...)	A timestamp or a unique ID per event
customer_type (new, returning, VIP)	Full URLs with tracking codes
button_location (header, floating, product)	Free text typed by visitors
delivery_area (9 districts)	Email addresses (also forbidden)

A dimension with thousands of different values ("high cardinality") fills reports with "(other)" rows and wastes a slot. Keep IDs for BigQuery (Section 21), not for custom dimensions.

SECTION 10

Consent, privacy and the Mauritius Data Protection Act 2017

Analytics data is about people. Cookie identifiers, IP addresses and user IDs can count as personal data when they can be linked to a person. Advanced tracking must therefore respect the law and your visitors' choices. This section is practical guidance, not legal advice.

What applies

- **Mauritius Data Protection Act 2017 (DPA 2017):** applies to processing of personal data in Mauritius. Key principles: lawful and transparent processing, collect only what you need, keep it secure, keep it no longer than necessary. Week 10 (Section 4) covers consent, registration with the Data Protection Commissioner and penalties.
- **GDPR:** can apply when you target or monitor people in the European Union, for example French or German tourists browsing a hotel website, or shoppers in Réunion.
- **Google's own policies:** no personally identifiable information in GA4; a privacy notice that explains your use of Google Analytics; and, for visitors in the European Economic Area (EEA), consent signals must be passed to Google for ads features.

Consent mode in plain English

A cookie banner (consent management platform, CMP) asks visitors whether they accept analytics and advertising cookies. **Consent mode** passes their choice to Google tags through settings such as **analytics_storage** and **ad_storage** (plus ad_user_data and ad_personalization in consent mode v2).

Implementation	What happens when a visitor refuses	Effect on data
Basic	Google tags do not load at all until consent is given	No data from refusing visitors; no modelling
Advanced	Tags load and send cookieless pings without identifiers	GA4 can estimate missing behaviour (modelled data) if the property is eligible

Behavioural modelling: when GA4 fills the gaps

With advanced consent mode, GA4 can use machine learning to estimate the behaviour of visitors who refused cookies, but only when the property has enough data. Google's stated prerequisites include at least 1,000 events per day with analytics_storage denied for at least 7 days, and at least 1,000 daily users with analytics_storage granted on at least 7 of the previous 28 days. You also need the **Blended** reporting identity (Admin → Data display → Reporting identity). Most small Mauritian sites will not reach these thresholds, so expect refused consent to mean missing data, not modelled data.

Source: Google Analytics Help, "Behavioral modeling for consent mode", checked October 2026.

Privacy checklist for analytics

- ☐ Privacy notice explains GA4, cookies, data retention and how to object
- ☐ Cookie banner with real choices where required (EU visitors) and consent mode configured in GTM
- ☐ No names, emails or phone numbers in URLs, events or parameters
- ☐ Data retention set deliberately (Section 27), Google signals switched on only if your notice covers it
- ☐ Access to GA4, GTM, Data Studio and BigQuery limited to people who need it, with two-factor authentication
- ☐ Business registered with the Data Protection Commissioner if required (Week 10)

PART C

Analyse in GA4

Custom reports, comparisons, explorations, audiences and channel groups

SECTION 11

Customising GA4 reports: library, comparisons and filters

Standard reports are not fixed. You can add dimensions and metrics, save your own versions, build collections in the menu and compare groups side by side. Customised reports are great for things you check every week; explorations (Sections 12–15) are for one-off questions.

Customise a report

1. Open a report, for example Reports → Acquisition → Traffic acquisition.
2. Click the pencil icon (Customise report) at the top right. You need the Editor role or above.
3. Under **Dimensions**, add the dimensions you want and choose the default one (for example Session default channel group, then add Landing page and Device category).
4. Under **Metrics**, add, remove and reorder metrics: Sessions, Engagement rate, Key events, Session key event rate, Total revenue.
5. Under **Charts**, choose which charts appear. Under **Report filter**, you can lock the report to, say, Country = Mauritius.
6. **Save as a new report** (do not overwrite the original) with a clear name: "Channels – key events (LG)".

Add it to the menu with the Library

Reports → **Library** (at the bottom of the left menu) lets you create a collection: a group of topics and reports shown in the left menu. Build a collection called "Lakaz Gato weekly" with your 4–5 customised reports, then publish it. Everyone with access now sees the same reports in the same place. The Library is also where you publish the Search Console collection after linking (Section 21).

Comparisons

Click **Add comparison** at the top of any report to show several groups side by side, for example "Device = mobile" vs "Device = desktop", or "Country = Mauritius" vs "Country = France". Every chart and row splits by the groups. Comparisons are the fastest way to apply the rule "segment before you conclude".

Question	Comparison to add
Is mobile performing worse?	Device category = mobile vs desktop
Do locals and tourists behave differently?	Country = Mauritius vs Country ≠ Mauritius
Is our paid traffic good quality?	Default channel group = Paid Social vs Organic Social
Does a campaign bring new people?	New / established = new vs established

Filters, date comparison and annotations

- **Date range → Compare:** previous period or same period last year. Use "same period last year" for seasonal businesses (Divali, Christmas, wedding season, school holidays).
- **Search box and filters** in tables: type "wedding" to filter landing pages, or click a row to see its detail.
- **Annotations:** add a note on the chart date when you launched a campaign, changed the website or fixed tracking. Three months later, nobody remembers why traffic jumped.

> **Tip:** customised reports and comparisons use aggregated data, so they are not limited by the data retention setting. Explorations are (Section 27). For year-on-year comparisons, use standard reports or Data Studio.

SECTION 12

Explorations: the free form technique

Explore (left menu) opens GA4's analysis workspace. You choose a technique, drag dimensions and metrics, apply segments and filters, and build tables or charts that standard reports cannot produce.

Seven exploration techniques and the question each answers

Explore → Blank, or pick a template. Explorations use the data retention period you set (2 or 14 months).

FREE FORM

Free form

Any table or chart with your own dimensions, metrics, segments and filters.

Key event rate by landing page × device

FUNNEL

Funnel exploration

Steps to a goal and where people drop off. Open or closed, with elapsed time.

Product → cart → checkout → purchase

PATH

Path exploration

What people do before or after an event or page.

What happens after the delivery page?

SEGMENT OVERLAP

Segment overlap

How up to 3 segments overlap.

Mobile + paid social + purchasers

COHORT

Cohort exploration

Do users who started in the same week come back?

Week 1 retention by acquisition week

USER LIFETIME

User lifetime

Value and behaviour of users over their whole lifetime.

Lifetime revenue by first user source

The workspace

Panel	What it holds
Variables (left)	Date range, segments, dimensions and metrics you import for this exploration
Settings (middle)	The technique, visualisation, rows, columns, values, breakdowns and filters
Canvas (right)	The result. Right-click a row to include, exclude, or create a segment or audience from it

Build a free form exploration step by step

Question: "Which landing pages turn mobile visitors from social media into orders?"

1. Explore → **Blank**. Name it "Landing pages × device – key event rate".
2. Date range: last 90 days. Dimensions (+): Landing page + query string, Device category, Session default channel group. Metrics (+): Sessions, Engagement rate, Key events, Session key event rate.
3. Drag **Landing page** to Rows, **Device category** to Columns, and the metrics to Values.
4. Filter: Session default channel group **exactly matches** Organic Social (or Paid Social). Show rows: 25.
5. Cell type: **Heat map** to make high and low values stand out.

6. Read it: which pages have many sessions but a low key event rate on mobile? That is your priority list.

Useful limits and habits

- Each user can create up to 200 explorations per property, and up to 500 can be shared per property. Delete old tests.
- Free form and funnel explorations accept up to 4 segments to compare; segment overlap accepts 3.
- On standard properties, explorations that query more than 10 million events are **sampled** (estimated from part of the data). The shield or quality icon at the top right shows if sampling or thresholds apply. Most small Mauritian sites never reach this.
- Explorations are private by default. Click **Share** to make a read-only copy visible to others in the property.
- Export to Google Sheets, CSV, Excel or PDF from the top right menu, for example to continue the analysis in Excel.

Source: Google Analytics Help on explorations, as summarised by Relevant Audience, checked October 2026.

SECTION 13

Funnel exploration, the advanced way

In Week 2 you built a simple funnel. The funnel exploration has more options that turn it from a picture into a diagnosis: open vs closed funnels, "directly followed by", elapsed time, next action and breakdowns.

Settings that change the answer

Setting	What it does	When to use it
Open vs closed funnel	Closed: users must enter at step 1. Open: users can join at any step	Closed for a strict checkout; open for a website where people land on product pages directly
Indirectly vs directly followed by	Indirectly: other events may happen in between. Directly: the next step must be the very next event	Keep "indirectly" unless you test a strict sequence
Within (time limit)	Step must happen within a set time after the previous one	"add_to_cart then purchase within 30 minutes"
Show elapsed time	Average time between steps	Find slow steps (delivery form, payment page)
Make trended funnel	Shows each step over time instead of bars	Check if a change on a date improved completion
Breakdown	Splits each step by a dimension (up to a few values)	Device category, channel, first-time vs returning
Next action	Shows what people did after leaving a step	Did they go to the delivery page, the FAQ, or WhatsApp?

From funnel to opportunity sizing

A drop-off only matters if fixing it is worth money. Use this simple calculation (you will practise it in the Excel exercise):

> **Extra orders** = users who reach the weak step × (target completion rate − current completion rate) × completion rate of all following steps.
> **Extra revenue** = extra orders × average order value.

Example (illustrative): 1,200 mobile users begin checkout each month and 38% complete payment. Desktop completes at 55%. If mobile reached 50%, that is 1,200 × (0.50 − 0.38) = 144 extra completed checkouts. With an average order of Rs 2,400, the opportunity is about Rs 345,600 a month. That number justifies fixing the mobile delivery form before redesigning the home page.

> **Tip:** always build the same funnel broken down by **Device category** first. On Mauritian sites the mobile funnel is usually where the money is lost.

SECTION 14

Path exploration and segment overlap

Path exploration shows the routes people take, and segment overlap shows how groups of users intersect. Both are good at revealing behaviour you did not expect.

Path exploration

1. Explore → **Path exploration**. Click "Start over".
2. Choose a **starting point** (an event or page, for example page_view on /cakes/wedding) or an **ending point** (for example generate_lead) to see what happened before.
3. Switch the node type between **Event name** and **Page title and screen name** (or page path).
4. Click a node to expand the next step. Look for loops (people going back and forth between two pages) and unexpected exits.

Question	Starting / ending point	What to look for
What do people do after viewing the prices page?	Start: page_view /prices	Do they go to order, WhatsApp, or leave?
What happens before a quote request?	End: generate_lead	Which pages convince: gallery, reviews, FAQ?
Where do blog readers go next?	Start: page_view /blog/...	Do they reach a product page at all?

Segment overlap

Segment overlap shows up to three segments in a Venn diagram, with the number of users in each intersection. Example: "Mobile users", "Paid Social sessions" and "Purchasers". If very few users sit in the centre, paid social brings mobile visitors who rarely buy. Right-click any intersection to create a new segment or an audience from it, for example "mobile paid social visitors who did not purchase" for a remarketing campaign (Section 16).

User explorer

User explorer lists anonymous app-instance or client IDs and lets you follow the event stream of one user. It is useful to debug tracking ("why does this user have two purchases?"), not to spy on individuals. Never try to identify a real person from it.

SECTION 15

Cohort exploration and user lifetime

Cohorts answer the question "do people come back?". A cohort is a group of users who share a starting point, such as their first visit in the same week. You then watch what share of each group returns in the following weeks.

Build a cohort exploration

1. Explore → **Cohort exploration**.
2. Cohort inclusion: **First touch** (first visit), or Any transaction for buyers.
3. Return criteria: **Any event** (any return), or a key event such as purchase for repeat buyers.
4. Cohort granularity: **Weekly** for most small sites (daily is too noisy, monthly too slow).
5. Calculation: Standard (active in that week), Rolling (active in that week and every previous week) or Cumulative (adds up a metric such as revenue).
6. Breakdown: First user default channel group, to see which channels bring people who come back.

How to read a cohort table

Cohort (first visit week)	Users	Week 0	Week 1	Week 2	Week 3
1–7 Sep	820	100%	9.4%	6.1%	4.9%
8–14 Sep	760	100%	11.2%	7.0%	5.5%
15–21 Sep	940	100%	6.0%	3.8%	–
22–28 Sep	880	100%	12.5%	–	–

Illustrative data. Read across a row to follow one cohort over time; read down a column to compare cohorts at the same age.

- Week 0 is always 100%: everyone in the cohort was active in their first week.
- The 15–21 September cohort was big but returned less (6.0% in week 1). What happened that week? A broad paid campaign with a giveaway brought curious visitors, not future customers.
- The 22–28 September cohort returns best so far: the newsletter launch and a recipe series gave people a reason to come back.

User lifetime

The user lifetime technique shows lifetime metrics for users first acquired in your date range: lifetime revenue, lifetime engagement, lifetime transactions, and predictive metrics (purchase probability, churn probability) when the property is eligible. Use it with **First user source / medium** to answer "which channel brings the most valuable customers over time, not just the most first orders?". It needs time and volume: a new or small property will show thin data.

> **In Mauritius:** for gyms, guesthouses and cake shops, repeat customers are the profit. A cohort table by channel often shows that Google Business Profile and referrals bring fewer but more loyal customers than broad social giveaways.

SECTION 16

Audiences and segments

Segments and audiences look similar but do different jobs. A segment is a lens for analysis inside an exploration. An audience is a group GA4 keeps updating, which you can use in reports and send to Google Ads for remarketing.

	Segment	Audience
Where	Inside one exploration	Admin → Data display → Audiences; used across reports
Data	Applies to past data in the date range	Starts collecting from creation (with some backfill for up to 30 days in many cases)
Types	User, session or event segments	User-based, with conditions, sequences and membership duration
Shared with Google Ads	No	Yes, if Google Ads is linked and personalised advertising is allowed
Typical use	"Compare mobile buyers with desktop buyers"	"Remarket to cart abandoners in the last 7 days"

Build an audience step by step

1. Admin → Data display → **Audiences** → New audience → Create a custom audience.
2. Name it clearly: "LG – Cart abandoners 7 days".
3. Include users when: event **add_to_cart**. Add a condition group to **exclude** users when event **purchase**.
4. Membership duration: 7 days (people leave the audience after 7 days without matching again).
5. Optional: set an **audience trigger** to log an event when someone joins, which you can mark as a key event.
6. Save. Check its size after a few days in Audiences or in the Google Ads shared library.

Useful audiences for a small business

Audience	Conditions	Use
Engaged non-buyers	2+ engaged sessions, no purchase or lead, 30 days	Remarketing with reviews and an offer
WhatsApp clickers	whatsapp_click in last 14 days	Exclude from acquisition ads; follow-up content
Wedding interest	page_view on /wedding pages, 60 days	Wedding tasting campaign
Recent purchasers	purchase in last 30 days	Exclude from discount ads; cross-sell
Predictive: likely purchasers	Purchase probability (if eligible)	Higher bids in Google Ads

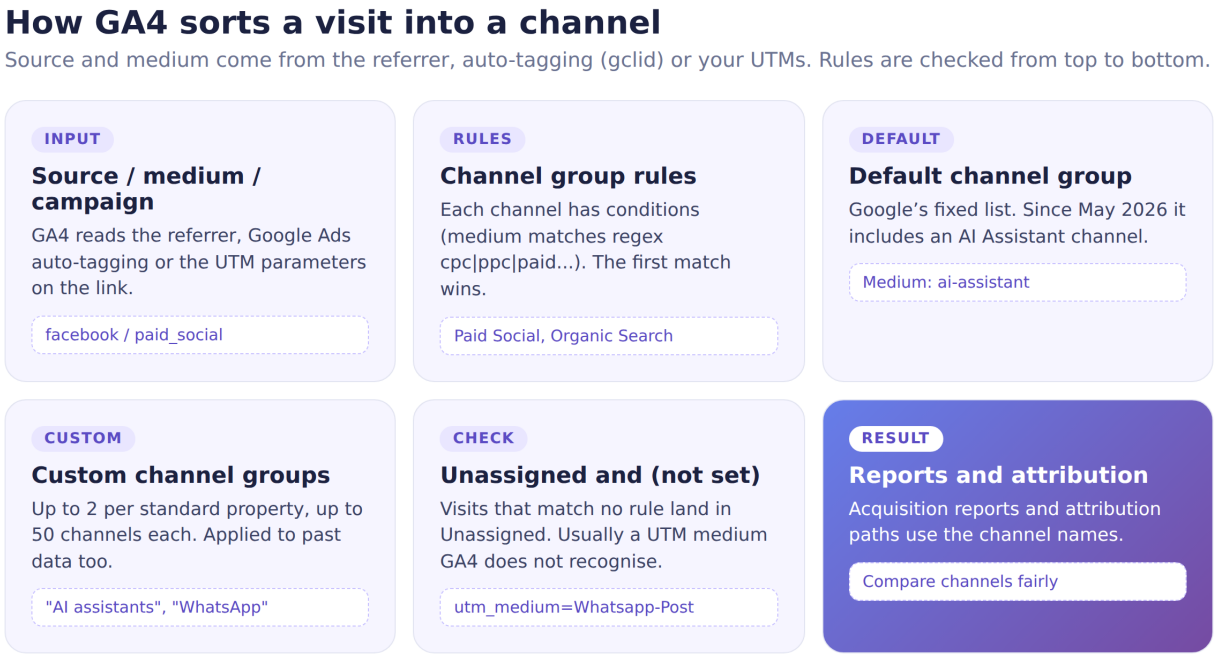
Predictive audiences need enough purchasers and churn data to train the model, so they often stay unavailable on small properties.

> **Tip:** in an exploration, right-click a row or a Venn intersection and choose "Create segment from selection" or "Build audience". It is the fastest way to turn an insight into an action.

SECTION 17

Channel groups: default, custom and AI assistants

Channel groups sort every visit into a channel such as Organic Search, Paid Social or Email. Getting this right matters, because acquisition reports, attribution and your dashboard all use these channels.



The default channel group and the new AI Assistant channel

Google maintains the default channel group and you cannot edit it. In May 2026 Google announced that the default group now includes an **AI Assistant** channel for visits referred by AI assistants; Google gave ChatGPT, Gemini and Claude as examples but did not publish a full list. These sessions show the medium value "ai-assistant". Before that, such visits usually appeared under Referral. Check your Traffic acquisition report to see whether the channel appears in your property.

Source: Google Analytics "What's new" notice of 13 May 2026, as reported by TechWyse, checked October 2026.

Create a custom channel group

1. Admin → Data display → **Channel groups** → Create new channel group. GA4 copies the default channels as a starting point.
2. Name it, for example "LG channels".
3. Add a channel: name "WhatsApp", condition Source **matches regex** whatsapp|wa\.me (your UTM source for WhatsApp links).
4. Add a channel "AI assistants" with Source matches regex such as chatgpt|openai|gemini|claude|perplexity|copilot. Google's own help article suggests a similar regex for custom AI channels.
5. Use **Reorder** to move your new channels **above Referral** (and above Organic Social for WhatsApp). Traffic is placed in the first channel whose rules it matches.

6. Save. Custom channel groups apply to past data too, so you can compare before and after at once.

Limits: up to 2 custom channel groups per standard property, with up to 50 channels in each (Google Analytics Help, checked October 2026). You can make a custom group the primary channel group so it is used in reports and attribution.

Fix "Unassigned" and messy channels

Problem	Cause	Fix
Lots of "Unassigned"	utm_medium not recognised (e.g. "Whatsapp-Post", "fb")	Use standard mediums: email, social, paid_social, cpc, referral, display, affiliate
Instagram ads in Organic Social	Ad links without UTMs or with utm_medium=social	utm_medium=paid_social (or cpc) on all paid links
Your own domain as a referral	Cross-domain journey (shop, booking engine, payment page)	Configure cross-domain measurement and unwanted referrals in the web stream
Newsletter visits in Direct	No UTMs on email links	UTMs on every email link (Week 10)

PART D

Attribution and beyond

Attribution models, conversion paths, their limits, product links, BigQuery and AI features in GA4

SECTION 18

Attribution in GA4: models and settings

Attribution decides how the credit for a key event is shared between the channels a customer used before converting. Week 2 introduced the idea. Here you learn exactly what GA4 offers in 2026, where to change it and what each choice does to your numbers.

One sale, three ways to give credit

Example journey for one Lakaz Gato order: Instagram ad → Google search → newsletter → purchase (Rs 3,000)

MODEL 1

Paid and organic last click

All the credit goes to the last channel clicked before the purchase (Direct is ignored).

Email: Rs 3,000

MODEL 2

Google paid channels last click

All credit to the last Google Ads click, if there is one; otherwise falls back to paid and organic last click.

No Google Ads click here: same as Model 1

MODEL 3 · DEFAULT

Data-driven

Credit shared by how much each touchpoint really raises the chance of buying, learned from your own data.

Illustrative: Paid Social 900 · Organic 1,200 · Email 900

The models available in 2026

Reporting attribution model	How it works	When it helps
Data-driven (default)	Uses your property's conversion data to estimate how much each touchpoint (clicks, and engaged views for YouTube) increased the chance of converting, then shares the credit accordingly	Default for most properties; fairer to channels that start or help journeys
Paid and organic last click	100% of the credit to the last channel the customer clicked through from before converting. Direct is ignored unless there is nothing else	Simple, stable, easy to explain; good for small properties with few conversions
Google paid channels last click	100% to the last Google Ads click in the path; if there is none, it behaves like paid and organic last click	When you mainly want to judge Google Ads

First click, linear, time decay and position-based models were removed from GA4 in November 2023. You will still see them in older articles and courses.

Where to change attribution settings

Admin → Data display → **Attribution settings** (you need the Marketer role or above). Google is gradually adding finer options, so the exact layout can differ in your property.

Setting	Options	Applies to
Reporting attribution model	Data-driven · Paid and organic last click · Google paid channels last click	Historical and future data
Lookback window: acquisition key events (first_visit, first_open)	30 days (default) or 7 days	Future data only

Setting	Options	Applies to
Lookback window: all other key events	90 days (default), 60 or 30 days	Future data only
Channels that can receive credit (Google Ads)	Paid and organic, or Google paid channels	Future data only

Source: Google Analytics Help "Select attribution settings" and Admin API documentation, checked October 2026.

Which numbers use attribution?

- **Attributed (uses the model):** key events and revenue in the Advertising section, and the key events / revenue columns in Traffic acquisition when you choose the key event dimension scope. This is where model choice changes the numbers.
- **Not attributed:** session counts by session source / medium simply show where each session came from. User acquisition uses the first user source.
- A lookback window of 90 days means a click up to 90 days before the key event can receive credit. For a cake order, decisions take days; for a wedding venue or a training course, weeks or months.

> **Tip:** for a small business with few key events a month, data-driven results can jump around. Look at **both** data-driven and paid and organic last click (Section 19) and focus on the direction of the difference, not on decimals.

SECTION 19

Reading attribution reports: paths and model comparison

GA4's Advertising section (left menu) holds the attribution reports. Two are essential: the attribution paths report, which shows the journeys, and the model comparison report, which shows how credit changes when you change the model.

Attribution paths (conversion paths)

Advertising → Attribution → **Attribution paths**. Choose a key event (for example purchase). The report splits each path into three phases:

- **Early touchpoints:** the first 25% of interactions in the path.
- **Mid touchpoints:** the middle 50%.
- **Late touchpoints:** the last 25%.

The table below the chart lists the most common paths, with key events, revenue, days to key event and touchpoints to key event. Look for channels that appear often early (they **start** journeys) and channels that appear late (they **close**).

Path (illustrative)	Key events	Days to key event
Organic Search	41	0.6
Paid Social → Organic Search	23	4.2
Paid Social → Email → Direct	14	9.8
Organic Social → Organic Search → Email	11	12.5
AI Assistant → Organic Search	4	2.1

Model comparison

Advertising → Attribution → **Model comparison** (Google also calls it the key event attribution models report). Pick two models, for example data-driven vs paid and organic last click, and a key event. The table shows key events and revenue for each channel under both models and the **% change**.

1. Sort by % change.
2. Channels with a big **positive** change under data-driven are **assist channels**: they help journeys that another channel closes. Typical: Paid Social, Display, Video, Organic Social.
3. Channels with a **negative** change are **closers**: they get a lot of last-click credit. Typical: Email, Organic Search for brand names, Direct (in other tools).
4. Before cutting a channel that looks weak on last click, check its share of early touchpoints and its change under data-driven.

> **How to write an attribution finding:** "Under last click, Paid Social gets 18 orders (Rs 41,000). Under data-driven it gets 29 orders (Rs 66,000), +61%. It appears early in 38% of purchase paths. Recommendation: keep the Paid Social budget, judge it on assisted value, and test a 4-week budget cut in one region before deciding."

SECTION 20

The limits of attribution, and what to do about them

Attribution reports look precise, but they only see what GA4 sees: clicks and engaged video views on tracked websites, from visitors who accepted cookies, on one device. In Mauritius, where WhatsApp and word of mouth close many sales, the blind spots are large.

Blind spot	Example	Mitigation
Views without clicks	Someone sees an Instagram Reel, later searches your name on Google	Watch branded search and Direct trends after campaigns; use platform reports for reach
WhatsApp, phone and shop sales	The order happens in a chat or at the counter	Track whatsapp_click as a key event; add a campaign code to the pre-filled WhatsApp text ("Hi! Code IG10"); log the source of each order
Refused cookies	Visitors who decline analytics cookies	Advanced consent mode; accept that totals are estimates (Section 10)
Cross-device journeys	Browse on phone, order on office laptop	User ID for signed-in customers; Google signals where allowed
Word of mouth and offline media	Radio, a flyer, a friend’s recommendation	"How did you hear about us?" on forms; QR codes with UTMs; vanity URLs (lakazgato.mu/radio)
Small numbers	15 orders a month	Use longer periods, combine with last click, avoid over-reading decimals
Platforms grading their own homework	Meta and Google Ads both claim the same sale	Use GA4 as the neutral referee; never add up platform conversions

Beyond attribution: testing what really works

Attribution shows correlation (who was in the journey), not causation (who made the sale happen). The more reliable way to know whether a channel adds sales is an **incrementality test**:

- **On/off test:** pause a channel for 2–4 weeks (or run it in only some regions) and compare orders with a similar period or region.
- **Holdout:** Meta and Google offer conversion lift studies for larger budgets; small businesses can do a simple version with a promo code only shown in one channel.
- **Marketing mix modelling (MMM):** statistical models such as Google's open-source Meridian estimate the effect of each channel from months of spend and sales data. They need volume and expertise, so they are for larger advertisers.

> **Do not treat attribution as the truth.** Treat it as one piece of evidence. A good recommendation combines attribution paths, platform data, business knowledge and, when the stakes are high, a test.

SECTION 21

Linking Search Console, Google Ads and BigQuery

GA4 becomes much more useful when it is connected to other Google products. All three links are free to set up from Admin → Product links.

Search Console

1. Admin → Product links → **Search Console links** → Link. You must be an owner of the Search Console property and an Editor in GA4.
2. Choose the Search Console property and the GA4 web stream.
3. Reports → **Library** → find the Search Console collection → Publish. Two reports appear: **Queries** (Google search terms, clicks, impressions, average position) and **Google organic search traffic** (landing pages with search and GA4 engagement data side by side).

Use it to find pages that rank and get impressions but few clicks (rewrite the title and description, Week 3), and pages that get search clicks but low engagement or key events (improve the content and call to action).

Google Ads

- Admin → Product links → **Google Ads links**. Enable personalised advertising and auto-tagging.
- Import GA4 key events into Google Ads as conversions (Weeks 6 and 7), so Smart Bidding optimises for real business results.
- Share GA4 audiences (Section 16) with Google Ads for remarketing.
- See cost, clicks and ROAS (return on ad spend) next to GA4 behaviour in Advertising reports.

BigQuery export: your raw data, free to start

BigQuery is Google Cloud's data warehouse. The GA4 link exports every raw event, with all parameters and user properties, into tables you query with SQL. Unlike explorations, your history is not limited by GA4's 14-month retention, and nothing is sampled or thresholded.

Fact (checked October 2026)	Detail
Cost of the link	Available to standard properties; you pay only BigQuery storage and query costs
Daily export	One table per day; standard properties have a daily limit of 1 million events
Streaming export	Near real time, no volume limit, costs about US$0.05 per GB (about 600,000 events per GB)
BigQuery sandbox	No credit card; 10 GB storage and 1 TB of queries a month free; tables expire after 60 days
History	Export starts from the day you link; data from before is not exported

Sources: Google Analytics Help "BigQuery Export"; BigQuery sandbox limits as documented by Google Cloud and DataCamp.

1. Create a Google Cloud project (console.cloud.google.com) and open BigQuery. Use the sandbox to start without billing.
2. In GA4: Admin → Product links → **BigQuery links** → Link → choose the project and the data location.
3. Choose **Daily** export (streaming needs billing) and the streams and events to include.
4. The next day, a dataset "analytics_<property ID>" appears with tables events_YYYYMMDD.

> **Tip:** even if you will not write SQL this year, link BigQuery now. It starts building your raw history from today. With the sandbox, tables expire after 60 days, so add billing (a few cents a month for a small site) when you want to keep the data.

SECTION 22

AI features in GA4: insights, anomalies and predictions

GA4 includes several AI and machine learning features. They save time and point you to changes you might miss, but they do not replace your judgement: always verify an AI answer in the reports before acting.

Feature	What it does	Where
Generated insights and anomaly detection	Automatically flags unusual changes ("Sessions from Paid Social up 140% yesterday") and summarises trends	Home page insight cards; Insights panel in reports
Custom insights	Alerts you set yourself: "email me if purchases drop more than 30% week on week"	Insights panel → View all insights → Create
Analytics Advisor (Gemini)	Answers questions in plain English with charts and explanations, suggests follow-up questions and checks your set-up	Rolling out since November 2025 to properties using the English interface
Predictive metrics	Purchase probability, churn probability and predicted revenue for users	Explorations and audiences, when eligible
Data-driven attribution and modelling	Shares credit and fills consent gaps with machine learning	Sections 10 and 18

Sources: Google Analytics Help; Loves Data review of Analytics Advisor (November 2025). At Google Marketing Live in May 2026 Google announced that the assistant is becoming a wider "Advisor" across Google products; availability varies by property.

Asking good questions to an AI assistant

- Be specific about the period, metric and comparison: "Compare session key event rate by device for September vs August 2026" works better than "how is my site doing?".
- Ask "why" questions as a starting point, then check the evidence: "Why did purchases fall last week?" → check the funnel and channel reports it points to.
- Ask configuration questions: "Is my data retention set to 14 months?", "Which events are key events?".
- Advisors can be wrong, can lack real-time or attribution data, and only see the property you are in. Confirm every number in a report.

Using general AI tools with your data

You can export a table (CSV) from an exploration and ask an AI assistant to spot patterns, write a summary for your manager or suggest hypotheses. Only share aggregated data, never personal data, and check every calculation it gives you in Excel. AI is excellent at the "Insight" step of Section 1 when you give it clean numbers; it is dangerous when it guesses the numbers.

PART E

Dashboards and data quality

Advanced Data Studio, the Analytics Report, audits, and why numbers differ between tools

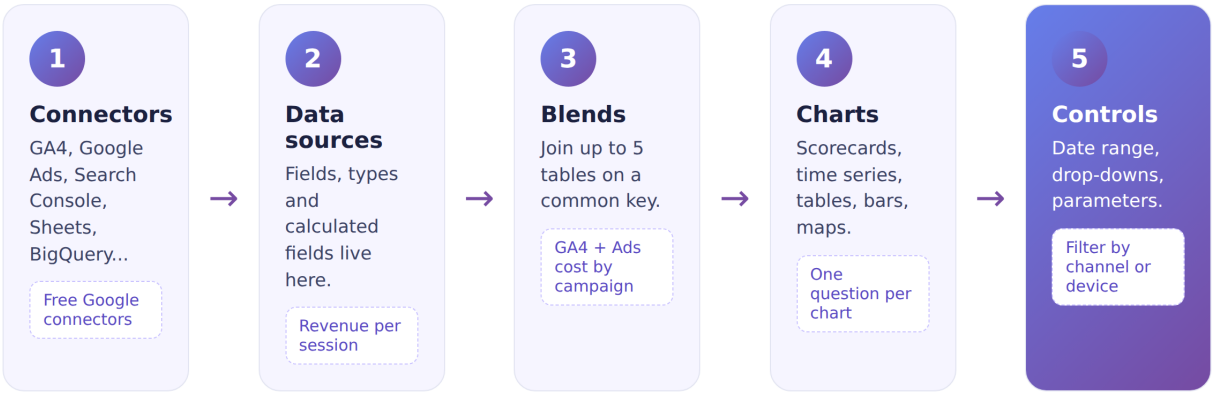
SECTION 23

Data Studio advanced: data sources, fields and calculated fields

In Week 2 you built a first Data Studio report. To build a dashboard people trust and use, you need to understand data sources, field types and calculated fields. Google brought back the Data Studio name in April 2026 (it was Looker Studio from 2022); existing reports and links moved over with no action needed.

How a Data Studio report is built

Data sources feed fields; fields feed charts; controls let viewers explore



Data Studio and Data Studio Pro

Data Studio is free for individuals and teams who build reports and dashboards. **Data Studio Pro** is a paid licence for organisations that adds AI features, team content management, deeper Google Cloud integration and enterprise security. Everything in this guide works in the free version.

Source: Google Cloud blog, "Looker Studio is now Data Studio", 10 April 2026.

Data sources

- A **data source** is the connection between a connector (GA4, Google Ads, Search Console, Google Sheets, BigQuery, MySQL, and hundreds of partner connectors) and your reports. It holds the list of fields, their types and your calculated fields.
- **Embedded** data sources live inside one report; **reusable** ones are saved separately and can feed many reports. Use reusable sources for your main GA4 property so a fix applies everywhere.
- **Credentials:** "Owner's credentials" lets viewers see data without their own GA4 access; "Viewer's credentials" requires each viewer to have access. Choose deliberately.
- **Data freshness:** GA4 data is cached (often 12 hours by default for GA4); you can change it in the data source settings or click Refresh data.

Calculated fields

In the data source (or in a chart, for a chart-only field), click **Add a field** and write a formula. Useful examples for GA4 data:

Field	Formula	Use
Revenue per session	Total revenue / Sessions	Compare channels on value, not volume
WhatsApp click rate	Event count for whatsapp_click / Sessions (build with a filtered metric or a blend)	Lead intent for WhatsApp-driven businesses
Channel (clean)	CASE WHEN REGEXP_MATCH(Session source, ".*(whatsapp|wa.me).*") THEN "WhatsApp" ELSE Session default channel group END	Report-level channel clean-up
Page type	CASE WHEN CONTAINS_TEXT(Landing page, "/blog") THEN "Blog" WHEN CONTAINS_TEXT(Landing page, "/cakes") THEN "Product" ELSE "Other" END	Group hundreds of pages into a few types
Market	IF(Country = "Mauritius", "Local", "International")	Locals vs tourists and diaspora

Ratios: divide totals, not averages. If you create "Revenue per session" as a calculated field, Data Studio divides the totals for each row, which is correct. If you average a ratio column across rows, small rows count as much as big ones and the result is wrong. The same rule applies in the Excel exercise.

SECTION 24

Blending, filters, controls and parameters

These four features turn a static report into a tool people can explore: blending joins data from different sources, filters and controls let viewers slice the data, and parameters let them change values in formulas.

Blending data

A blend joins up to **five** tables (data sources) on common fields, called join keys. Typical blends: GA4 sessions and key events + Google Ads cost by campaign; GA4 landing pages + Search Console impressions by page; GA4 orders by date + a Google Sheet of shop sales by date.

1. Select a chart → Data panel → **Blend data** (or Resource → Manage blends).
2. Add the second table and choose the join keys, for example Date and Session campaign = Campaign.
3. Choose the join type: **Left outer** (all rows from the left table, matches from the right) is the safe default; Inner keeps only matches; Right outer, Full outer and Cross join exist for special cases.
4. Add the dimensions and metrics you need from each table, then build calculated fields on the blend, for example Cost per key event = Cost / Key events.

Join keys must match exactly (Data Studio only supports equality). "Fete-des-meres" and "fete-des-meres" will not join, which is another reason for strict lowercase UTMs. Blends live inside the report where you created them.

Filters and controls

Feature	What it does	Example
Chart filter	Applies to one chart, set by the editor	Top pages table shows only /cakes pages
Page or report filter	Applies to all charts on a page or in the report	Exclude internal test hostname
Date range control	Viewer chooses the period	Default "Last 28 days" with comparison to previous period
Drop-down control	Viewer filters by a dimension	Channel, Device category, Market
Advanced filter / input box	Viewer types text to filter	Search a landing page
Data control	Viewer switches between accounts or properties	Agency reports for several clients

Parameters

A parameter is a value the viewer can change, used inside calculated fields. Example: a "Target key event rate" parameter (default 3%) used in a field **Gap to target = Session key event rate − Target**. Add a slider or input box control linked to the parameter, and the manager can test "what if the target were 4%?". Another classic: a "Margin %" parameter to turn revenue into estimated gross profit.

Templates, themes and sharing

- **Theme and layout:** set your brand colours and fonts once (Theme and layout), use a grid and the same chart styles on every page.
- **Templates:** start from the Data Studio template gallery or copy a good report (File → Make a copy) and swap the data source. Build your own template for repeat clients.
- **Sharing:** Share → invite people as viewers or editors, or get a link. Schedule email delivery (for example every Monday at 8:00). Check the data source credentials before sharing outside the company.
- **Ownership:** create important reports with a business Google account, not a personal one, and add a second owner, so the report survives if someone leaves.

SECTION 25

Designing your Analytics Report dashboard

Your deliverable is a custom GA4 dashboard in Data Studio with at least 5 metrics and an attribution analysis. A good dashboard answers the questions of one audience in under a minute. Design it on paper before you build it.

Recommended layout: three pages

Page	Content	Charts
1 · Overview	The 5–7 KPIs against the previous period and target; trend over time; channel summary	Scorecards with comparison; time series; bar chart by channel
2 · Acquisition and attribution	Channels by sessions, engagement, key events, revenue per session; AI Assistant and WhatsApp channels; attribution comparison	Table with heat map; bar chart last click vs data-driven (from a Sheet or GA4 export)
3 · Behaviour and conversion	Landing pages, device split, funnel steps, top products or services	Table; stacked bar by device; funnel numbers table

Choosing the 5+ metrics

KPI	GA4 field	Why it belongs on the dashboard
Sessions and Total users	Sessions, Total users	Reach and returning visits
Engagement rate	Engagement rate	Quality of traffic
Key events	Key events (filtered on the main key event)	Business results
Session key event rate	Session key event rate	Conversion efficiency
Revenue and revenue per session	Total revenue; calculated field	Value, not just volume
WhatsApp clicks (or leads)	Event count, event name = whatsapp_click	Leads that close off-site
New vs returning	New users, Returning users	Loyalty

Design rules

1. **One audience, one purpose.** The owner wants "are we growing and where should we spend?". The web developer wants page speed and errors. Do not mix.
2. **Most important at the top left.** Scorecards first, detail lower down.
3. **Always show a comparison** (previous period or target) and use green and red sparingly.
4. **Right chart for the question:** time series for trends, bars for comparing categories, tables for detail, scorecards for headline numbers. Avoid pies with more than 4 slices and 3D anything.
5. **Label everything:** title with the question it answers, a date range control, a short note on definitions ("Key events = orders + WhatsApp clicks").
6. **Add an insight box** on page 1: 3 bullet points written by a human each month: what happened, why, what we will do.

In Mauritius: many owners read reports on WhatsApp on their phones. Schedule a PDF delivery, keep page 1 readable on a small screen, and send a 3-line summary with the link. If the dashboard is not opened, it does not exist.

SECTION 26

The data quality audit

A beautiful dashboard built on broken tracking is worse than no dashboard, because people trust it. Run this audit before you build your Analytics Report, and every quarter after that.

The analytics data quality audit

Do it before you trust a dashboard, and every quarter after that

1 · SETTINGS

Property basics

Time zone Mauritius, currency MUR, 14-month retention, internal traffic filter active, unwanted referrals listed.

Admin checks

2 · TAGS

Tagging

Google tag fires once per page; no double tagging; GTM versions named; consent mode set up.

Tag Assistant, Preview

3 · EVENTS

Events and key events

Every planned event fires with its parameters; key events not duplicated; purchase values correct.

DebugView, test order

4 · SOURCES

Traffic sources

Low share of (not set) and Unassigned; UTMs follow the naming rules; no self-referrals.

Traffic acquisition

5 · DEFINITIONS

Custom definitions

Needed parameters registered; no high-cardinality dimensions; names documented.

Admin → Custom definitions

6 · RECONCILE

Compare with the till

GA4 revenue and orders vs the shop system or CRM; explain the gap.

Aim for a stable gap

Audit checklist with where to look

Check	Where	Good result
Time zone and currency	Admin → Property details	Mauritius time, MUR
Data retention	Admin → Data collection and modification → Data retention	14 months
Internal traffic filter	Admin → Data filters	Active, with current office IPs
Unwanted referrals	Web stream → Configure tag settings → List unwanted referrals	Payment gateways and booking engines listed
Tag fires once	Tag Assistant on 3–4 page types	One Google tag, one page_view per page
Key events	Admin → Key events; DebugView with a test	Each fires once, at the right moment
Purchase data	Test order; Monetization reports	Value in MUR, unique transaction_id, items present
(not set) and Unassigned	Traffic acquisition; Landing page report	Small share, explained
UTM consistency	Session campaign / source / medium tables	Lowercase, standard mediums, no duplicates
Custom definitions	Admin → Custom definitions	Every parameter in the plan registered
Consent	Tag Assistant consent tab; banner test	Choices respected; consent mode signals sent
Access	Admin → Account / property access management	Only current people; at least two administrators

Reconcile with the business

Compare GA4 with your "source of truth" (shop system, booking engine, CRM or sales log) for the same month. GA4 is almost always lower, because of refused cookies, ad blockers and orders taken offline. That is fine if the gap is **stable and explained**. A sudden change in the gap means tracking broke or a process changed.

Month (illustrative)	Shop system orders	GA4 purchases	GA4 ÷ shop
July	212	171	81%
August	238	195	82%
September	251	139	55%: investigate

In this example the drop in September matched the date a new payment page was installed: customers returned from the payment gateway as a "referral" and some purchase events were lost. Fix: list the gateway as an unwanted referral and test the confirmation page.

SECTION 27

Why numbers differ: retention, thresholds, sampling and modelled data

"GA4 says 1,250 users, Data Studio says 1,190 and Meta says 1,800 clicks!" Different numbers are normal. An analyst knows the reasons and explains them calmly.

Reason	What happens	What to do
Data retention	Explorations and funnel reports only use data within the retention period (2 or 14 months on standard properties). Standard aggregated reports are not affected. Age, gender and interest data is kept 2 months	Set 14 months on day one; use standard reports, Data Studio or BigQuery for longer history
Data thresholds	GA4 hides rows with few users to protect privacy, mainly when Google signals or demographic data is involved. A shield icon says "Data thresholds applied"	Wider date range; fewer dimensions; Reporting identity Device-based if your privacy notice allows; BigQuery
Sampling	Explorations above 10 million events (standard) are estimated from a sample	Shorter ranges; standard reports; BigQuery
Modelled data	With advanced consent mode and an eligible property, GA4 estimates refusing visitors	Know which reporting identity is used (Blended includes modelling)
Cardinality "(other)"	Tables with many unique values group the rest into (other)	Fewer high-cardinality dimensions; explorations; BigQuery
Clicks vs sessions	Ad platforms count clicks; GA4 counts sessions. Bounced clicks, blocked tags and slow pages create gaps	Compare trends, not exact numbers; check landing page speed
Attribution and dates	Platforms credit conversions to the click date and their own model; GA4 to its model and the key event date	State the model and date logic on every report

Sources: Google Analytics Help "Data retention" and "Behavioral modeling for consent mode"; thresholding as explained by Analytify, checked October 2026.

Reporting identity in one minute

Admin → Data display → **Reporting identity** decides how GA4 recognises users: **Blended** (User ID, then device ID, then modelling), **Observed** (User ID, then device ID, no modelling) or **Device-based** (device ID only). It changes how users are counted in reports, not the data collected, and you can switch back. Device-based often removes thresholds, but you lose cross-device and modelled data.

> **Tip:** add a "Data notes" box to your dashboard: retention setting, attribution model, reporting identity, known tracking issues and the date of the last audit. It prevents most "the numbers are wrong" discussions.

PART F

Put it to work

A case study, your Analytics Report, templates, checklists and glossary

SECTION 28

Case study: Lakaz Gato finds its missing orders

"Lakaz Gato" is a fictional but realistic cake shop in Rose Hill that you met in Week 2. After building a website and a first dashboard, it sells online (card payment and cash on delivery) and through WhatsApp. In September 2026 online orders fell and the owner wanted to cut the Instagram ads budget. All numbers are illustrative.

1. The question and hypotheses

Question: why did online orders fall from 238 in August to 201 in September, and should we cut Instagram ads?

Hypotheses: (a) Instagram ads bring poor-quality traffic; (b) something on the website changed; (c) orders moved to WhatsApp.

2. What the team did (two weeks)

- **Audit (Section 26):** found that the new payment page appeared as a referral and that 14% of sessions had utm_medium "Instagram" (not recognised, so Unassigned). Fixed both and documented the change.
- **Tracking (Sections 6–9):** added order_form_submit with cake_type and delivery_area from the dataLayer, whatsapp_click with button_location, and registered four custom dimensions.
- **Channels (Section 17):** custom channel group with "WhatsApp" and "AI assistants" channels; fixed UTM mediums to paid_social.
- **Funnel (Section 13):** checkout funnel by device for August vs September.
- **Attribution (Sections 18–19):** model comparison for purchase and whatsapp_click, data-driven vs paid and organic last click, and attribution paths.
- **Dashboard (Section 25):** three-page Data Studio report shared with the owner every Monday.

3. What the data said

Finding	Evidence
The drop was mobile-only	Mobile begin_checkout → purchase fell from 52% to 36% after the new delivery form on 2 September; desktop unchanged at 58%
Instagram ads start journeys	Paid Social: 19 purchases on last click vs 31 on data-driven (+63%); early touchpoint in 41% of purchase paths
WhatsApp intent grew	whatsapp_click +28%, mostly from the delivery page: people gave up on the form and asked on WhatsApp
AI assistants are small but valuable	96 sessions in September with a 6.3% key event rate, double the site average

4. Decisions and results (illustrative)

Decision	Result after 6 weeks
Rebuild the mobile delivery form (3 fields fewer, district drop-down)	Mobile checkout completion back to 51%; about 40 extra orders a month
Keep the Instagram budget; judge it on data-driven key events and a 3-week regional test	Orders fell 18% in the test region with ads paused: ads confirmed as incremental
Pre-filled WhatsApp message with a code ("Hi! I saw the IG cake - code IG")	WhatsApp orders now logged with their source in the sales sheet
Improve the FAQ and prices page that AI assistants quote	AI Assistant sessions doubled (still small)

> **The lesson:** the first idea ("Instagram doesn't work, cut it") was wrong. A clean audit, a funnel by device and an attribution comparison found the real problem (the mobile form) and protected a channel that was starting journeys. Questions first, then the right exploration, then a decision with a test.

SECTION 29

Exercises and your Analytics Report

Your Week 11 deliverable is an **Analytics Report**: a custom GA4 dashboard in Data Studio with at least 5 metrics, and an attribution analysis with recommendations. Use your own business, your employer's (with permission) or the free GA4 demo account (Google Merchandise Store).

Exercise A · Measurement plan and custom events (60 min)

Write a measurement plan for one business (Section 5): 2 objectives, 4–6 KPIs, 5–8 events with parameters. Build at least one custom event with parameters in GTM (Section 7), test it in DebugView and register one custom dimension (Section 9). Screenshot Tag Assistant and the custom definition.

Exercise B · Explorations and channels (60 min)

In your property or the demo account, build: a free form exploration (landing page × device), a funnel exploration with a device breakdown, and a cohort exploration by week (Sections 12–15). Create one audience (Section 16) and one custom channel group with an "AI assistants" channel (Section 17). Write one insight and one decision for each exploration.

Exercise C · Excel exercise (90 min)

Download the Week 11 Excel file: GA4 export by channel and landing page with a pivot table, attribution comparison, funnel opportunity sizing, cohort retention and a dashboard specification. Scored out of 20.

Deliverable: the Analytics Report

Part	Content
1. Data Studio dashboard	Link to a report connected to GA4 (yours or the demo account): at least 5 metrics, 2–3 pages, a date range control and at least one other control, a comparison on every scorecard, one calculated field
2. Data notes	Property, date range, attribution model, retention, reporting identity, known tracking limits
3. Attribution analysis	Model comparison (data-driven vs paid and organic last click) for one key event; top 5 attribution paths; which channels start and which close journeys
4. Three insights	Each written as question → evidence (chart or number) → insight
5. Recommendations	3 decisions with owner, date, expected impact and how you will measure them; at least one test
6. Data quality	Results of a short audit (Section 26) and what you would fix first

Submit the Data Studio link (shared as "anyone with the link can view") plus a 2–4 page document with parts 2–6 on the learning platform, Module 11 exercises.

TEMPLATES

Copy-and-use templates

Measurement plan

Objective	KPI and target	Event(s)	Parameters	Dimension(s)	Owner / tested

Dashboard specification

#	KPI / question	GA4 field or formula	Chart	Comparison	Filter / control
1					
2					
3					
4					
5					

Insight card

Question: [what the business needs to know]
Hypothesis: [what we think is happening]
Evidence: [report / exploration, date range, numbers]
Insight: [one sentence]
Decision: [action] · **Owner:** [name] · **Date:** [when] · **Measure:** [KPI and target]

dataLayer brief for a developer

```
When the [form name] is submitted successfully (after validation), please push:
window.dataLayer = window.dataLayer || [];
window.dataLayer.push({
  event: '[event_name]',
  form_name: '[value]',
  [parameter_2]: '[value]'
});
Do not include names, emails or phone numbers. Test page: [URL]
```

SECTION 30

Checklists, mistakes and glossary

Advanced set-up checklist

- ☐ Measurement plan written and shared; naming rules agreed
- ☐ Custom events with parameters tested in Preview and DebugView
- ☐ Recommended e-commerce or lead events with value and MUR currency
- ☐ Custom dimensions registered; quota checked
- ☐ Custom channel group with WhatsApp and AI assistants channels, above Referral
- ☐ Attribution settings and lookback windows chosen and documented
- ☐ Search Console, Google Ads and BigQuery linked
- ☐ Consent banner and consent mode tested; privacy notice updated

Before you share a dashboard

- ☐ Data quality audit done; known issues listed in Data notes
- ☐ Every scorecard has a comparison; every chart title is a question
- ☐ Date range control and at least one filter control
- ☐ Ratios calculated from totals, not averaged
- ☐ Credentials and sharing checked; second owner added
- ☐ An insight box with 3 bullet points and decisions

Mistakes to avoid

Mistake	Do this instead
Tracking everything "just in case"	Measurement plan first, 5–10 well-defined events
Sending parameters but never registering them	Register custom dimensions on day one (not retroactive)
Inventing names for e-commerce events	Use Google’s recommended events and parameters
Comparing different scopes and blaming GA4	Match dimension and metric scopes
Cutting a channel on last-click numbers alone	Model comparison, paths and a test
Adding up conversions from Meta, Google Ads and GA4	One referee (GA4) and trends from each platform
Dashboards with 40 charts	5–7 KPIs, one audience, insights written by a human
Trusting an AI answer without checking	Verify every number in the reports

Glossary

Term	Meaning
Event / parameter	An interaction recorded by GA4 / a detail sent with it
User property	An attribute of a user stored with their events
Scope	The level a dimension or metric belongs to: user, session, event or item
Custom dimension / metric	A parameter registered so it can be used in reports
Calculated metric	A metric built with a formula from other metrics
dataLayer	A JavaScript list where the website sends messages to GTM
Exploration	GA4’s analysis workspace (free form, funnel, path, cohort...)
Segment / audience	A subset for analysis / a group of users kept up to date for reports and ads
Cohort	Users who share a start date, followed over time
Channel group	Rules that sort traffic into channels
Attribution model	The rule for sharing credit between touchpoints
Lookback window	How far back a touchpoint can receive credit
Data-driven attribution (DDA)	Credit shared using machine learning on your own data
Incrementality	The extra results a channel really causes, measured by a test
Thresholding / sampling	Hiding small rows for privacy / estimating from part of the data
Consent mode	Passing visitors’ cookie choices to Google tags
Blend	A join of up to five data sources in Data Studio
Parameter (Data Studio)	A value viewers can change that is used in calculated fields
BigQuery	Google Cloud data warehouse that receives raw GA4 events
Cardinality	Number of distinct values in a dimension